

ANSI SQL Using MySQL - Exercise Answers

1. User Upcoming Events

```
SELECT u.full_name, e.title, e.city, e.start_date FROM Users u JOIN Registrations r
ON u.user_id=r.user_id JOIN Events e ON r.event_id=e.event_id AND u.city=e.city
WHERE e.status='upcoming' ORDER BY e.start_date;
```

2. Top Rated Events

```
SELECT e.title, AVG(f.rating) avg_rating FROM Events e JOIN Feedback f ON
e.event_id=f.event_id GROUP BY e.event_id,e.title HAVING COUNT(*)>=10 ORDER BY
avg_rating DESC;
```

3. Inactive Users

```
SELECT * FROM Users u WHERE NOT EXISTS (SELECT 1 FROM Registrations r WHERE
r.user_id=u.user_id AND r.registration_date>=CURDATE()-INTERVAL 90 DAY);
```

4. Peak Session Hours

```
SELECT event_id, COUNT(*) total_sessions FROM Sessions WHERE TIME(start_time)
BETWEEN '10:00:00' AND '12:00:00' GROUP BY event_id;
```

5. Most Active Cities

```
SELECT u.city, COUNT(DISTINCT r.user_id) registrations FROM Users u JOIN
Registrations r ON u.user_id=r.user_id GROUP BY u.city ORDER BY registrations DESC
LIMIT 5;
```

6. Event Resource Summary

```
SELECT e.title, r.resource_type, COUNT(*) total FROM Events e LEFT JOIN Resources r
ON e.event_id=r.event_id GROUP BY e.title,r.resource_type;
```

7. Low Feedback Alerts

```
SELECT u.full_name,e.title,f.comments FROM Feedback f JOIN Users u ON
f.user_id=u.user_id JOIN Events e ON f.event_id=e.event_id WHERE f.rating<3;
```

8. Sessions per Upcoming Event

```
SELECT e.title, COUNT(s.session_id) session_count FROM Events e LEFT JOIN Sessions s
ON e.event_id=s.event_id WHERE e.status='upcoming' GROUP BY e.event_id;
```

9. Organizer Event Summary

```
SELECT u.full_name,e.status,COUNT(*) total_events FROM Users u JOIN Events e ON
u.user_id=e.organizer_id GROUP BY u.full_name,e.status;
```

10. Feedback Gap

```
SELECT e.title FROM Events e JOIN Registrations r ON e.event_id=r.event_id LEFT JOIN
Feedback f ON e.event_id=f.event_id GROUP BY e.event_id HAVING
COUNT(f.feedback_id)=0;
```

11. Daily New User Count

```
SELECT registration_date, COUNT(*) users_count FROM Users WHERE  
registration_date>=CURDATE()-INTERVAL 7 DAY GROUP BY registration_date;
```

12. Event with Maximum Sessions

```
SELECT e.title, COUNT(*) total_sessions FROM Events e JOIN Sessions s ON  
e.event_id=s.event_id GROUP BY e.event_id HAVING COUNT(*)=(SELECT MAX(cnt) FROM  
(SELECT COUNT(*) cnt FROM Sessions GROUP BY event_id)x);
```

13. Average Rating per City

```
SELECT e.city, AVG(f.rating) avg_rating FROM Events e JOIN Feedback f ON  
e.event_id=f.event_id GROUP BY e.city;
```

14. Most Registered Events

```
SELECT e.title, COUNT(*) registrations FROM Events e JOIN Registrations r ON  
e.event_id=r.event_id GROUP BY e.event_id ORDER BY registrations DESC LIMIT 3;
```

15. Event Session Time Conflict

```
SELECT s1.event_id,s1.title,s2.title FROM Sessions s1 JOIN Sessions s2 ON  
s1.event_id=s2.event_id AND s1.session_id>s2.start_time;
```

16. Unregistered Active Users

```
SELECT * FROM Users u WHERE u.registration_date>=CURDATE()-INTERVAL 30 DAY AND NOT  
EXISTS (SELECT 1 FROM Registrations r WHERE r.user_id=u.user_id);
```

17. Multi-Session Speakers

```
SELECT speaker_name, COUNT(*) sessions_handled FROM Sessions GROUP BY speaker_name  
HAVING COUNT(*)>1;
```

18. Resource Availability Check

```
SELECT e.title FROM Events e LEFT JOIN Resources r ON e.event_id=r.event_id WHERE  
r.resource_id IS NULL;
```

19. Completed Events with Feedback Summary

```
SELECT e.title, COUNT(DISTINCT r.registration_id) registrations, AVG(f.rating)  
avg_rating FROM Events e LEFT JOIN Registrations r ON e.event_id=r.event_id LEFT  
JOIN Feedback f ON e.event_id=f.event_id WHERE e.status='completed' GROUP BY  
e.event_id;
```

20. User Engagement Index

```
SELECT u.full_name, COUNT(DISTINCT r.event_id) attended_events, COUNT(DISTINCT  
f.feedback_id) feedbacks_submitted FROM Users u LEFT JOIN Registrations r ON  
u.user_id=r.user_id LEFT JOIN Feedback f ON u.user_id=f.user_id GROUP BY u.user_id;
```

21. Top Feedback Providers

```
SELECT u.full_name, COUNT(*) total_feedbacks FROM Users u JOIN Feedback f ON  
u.user_id=f.user_id GROUP BY u.user_id ORDER BY total_feedbacks DESC LIMIT 5;
```

22. Duplicate Registrations Check

```
SELECT user_id,event_id,COUNT(*) duplicates FROM Registrations GROUP BY  
user_id,event_id HAVING COUNT(*)>1;
```

23. Registration Trends

```
SELECT DATE_FORMAT(registration_date,'%Y-%m') month, COUNT(*) registrations FROM  
Registrations WHERE registration_date>=CURDATE()-INTERVAL 12 MONTH GROUP BY month  
ORDER BY month;
```

24. Average Session Duration per Event

```
SELECT e.title, AVG(TIMESTAMPDIFF(MINUTE,s.start_time,s.end_time)) avg_duration FROM  
Events e JOIN Sessions s ON e.event_id=s.event_id GROUP BY e.event_id;
```

25. Events Without Sessions

```
SELECT e.title FROM Events e LEFT JOIN Sessions s ON e.event_id=s.event_id WHERE  
s.session_id IS NULL;
```